CHAPTER 1

Getting Started

One of the keys to the Arduino's success is the minimal amount of time that it takes for a complete novice to move from opening the little box containing the Arduino interface board to having their first source code, also known as a sketch, up and running on that board.

The Arduino software development environment is free to download and use with no lengthy registration procedures or end-user agreements, and there is little to no setup to get the board running with your computer, regardless of the platform, working equally well on Mac, Linux, and Windows-based PCs. The Arduino web site at `www.arduino.cc` provides easy-to-follow "Getting Started" tutorials and whenever you get stuck there is always the active, vocal, and generally helpful Arduino community that is willing to share its knowledge. This low barrier to programming embedded electronics means it's possible to make lights blink in ten minutes flat with little to no prior experience.

This chapter will walk you through the history behind the Arduino platform, with an eye towards what makes the Arduino such a success story. We will look at some of the things that can be done with the platform and get you started on the right foot with a firm understanding of its possibilities. This includes a brief walk-through of that first ten-minute experience so that we can quickly move into some of our first projects in the next chapter. But first, let's start with a little background.

Arduino is for Makers

Arduino is a flexible programmable hardware platform designed for artists, designers, tinkerers, and the makers of things. Arduino's little, blue circuit board, mythically taking its name from a local pub in Italy, has in a very short time motivated a new generation of DIYers of all ages to make all manner of wild projects found anywhere from the hallowed grounds of our universities to the scorching desert sands of a particularly infamous yearly arts festival and just about everywhere in between. Usually these Arduino-based projects require little to no programming skills or knowledge of electronics theory, and more often than not, this handiness is simply picked up along the way.

Central to the Arduino interface board, shown in Figure 1-1, is an onboard microcontroller—think of it as a little computer on a chip.